

Objectives:

The objective of this lab session is to enable students to perform basic user management in a Linux environment. Students will learn about Root User in a Linux environment and the importance of sudo groups in the context of permissions and hierarchy of powers. Through hands-on practice, students will learn to create users. Students will be educated about the /root directory where the home directory of root user is maintained. Students will also learn about **/etc/passwd** file and fully understand all fields of a user. They'll be educated on system users which are created by the operating system to manage different utilities and processes. Students will also learn about groups and the difference between primary group and supplemental groups of a user. They'll also walkthrough **/etc/group** file.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- **PLO5: Modern Tool Usage**
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mapped PLOs
CL01	Practice Linux commands and utilities to gain proficiency with the command-line interface and use it effectively for system management tasks, basic scripting, software installation, and other essential operations in a Linux environment.	A5	PL05

Practice Exercises

Before we begin today's graded lab exercise:

- Open your **terminal** as we start the session.
- Follow along and replicate everything I do and try to internalize the commands and concepts.
- This will prepare you for today's **graded exercise**.

We practiced following commands in previous weeks ;

`echo, whoami, hostname, pwd, cd, ls, clear, script, touch, tee, mkdir`

`which, whatis, man, help, wc, cp, mv, head, tail, more, less, grep`

and explored their powers by using a lot of cool switches along with them and piping together multiple commands to perform multiple things as a single unit to do a lot of useful activities.

These are the commands we are planning to learn this week ;

`sudo, su, su - , adduser, groups, id`

and of course we'll also explore the use of different switches along with these commands for modifying their default behaviours and adding more functionality to them.

We will learn about the **root user** in linux and see where it maintains its directory and how we can execute commands with special privileges of the **sudo** group. How can we switch to the root user within our terminal session and how to exit from it.

We will learn how to create new users and verify it in the `/etc/passwd` file. We will also learn about primary group and supplemental groups of a user. We will learn about the `uid`, `gid` and other important information fields of a user record.

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Graded Exercises

Task

- Display a short introduction message containing your name and roll number on the terminal.
- Write some comments about root directory, root user, /home directory and root user's home directory. Explain the difference between them in your own words.
- Login to the root user account such that you land directly in the root user's home directory.
- Run whoami and pwd commands to verify this.
- Create a file in that directory and display the contents of this directory.
- Copy this file in the directory of your default user account without logging out of the root user and display the contents of that default user for verification.
- Create a new user account using adduser command and navigate to the /home directory to show that it's created successfully.
- From your /etc/passwd file fetch only that line which contains the newly added user and append it to a new file named after that user. You can create this file in the directory of your new user.
- Login to your new user such that you land directly in its home directory and print out the username and current working directory using whoami and pwd commands.

- Display the contents of this directory to show the file you created containing this user's entry in `/etc/passwd`.
- Display the contents of this file to show that information is gathered correctly.
- Try to copy this file in your default user's home directory and observe the output.
- Write some comments about your observations here.
- Print out information about the uid, gid and other supplemental groups names this user is part of.

Submission Guidelines

- Take screenshots of your work and submit a pdf document of those screenshots. You can use any online tool for creating pdf from these screenshots.

